

# Andy Huang

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## SUMMARY

Computer Science student at HKUST (graduating 2028) who ships complete systems end to end: a browser tool with 123 automated assertions in CI, a neural network quantised onto an 8-bit microcontroller, and hand-written C++ containers.

## EDUCATION

### The Hong Kong University of Science and Technology (HKUST)

2023 – 2028 (expected)

BSc in Computer Science

### Peking University — Globex Exchange Programme

Jul 2026

AI-Enabled Control Engineering

## EXPERIENCE

### SDALWINDOW — Part-Time Technical Assistant

Jan 2026 – Feb 2026

- Assisted licensed inspectors with aluminium window inspection, maintenance and small-scale replacement over a 30-day engagement.

### HKUST, School of Computer Science — Student Helper

Nov 2025

- Presented COMP programme structure, study pathways and career outcomes to students and parents.

### 4M Industrial Development Limited — Sales Assistant (Book Fair)

Jul 2024

- Sold and demonstrated science-education toys, handling customer enquiries.
- Supported booth setup and inventory tracking across six days of trading.

## PROJECTS

### AimSense — browser mouse-sensitivity testing tool

JavaScript · Canvas 2D

Zero dependencies · lyimou.github.io/aimsense · github.com/lyimou/aimsense

- Built and shipped a 3,985-line tool measuring FPS mouse sensitivity: three aim tests swept across 0.5× / 1.0× / 2.0×, then curve-fitted by least squares to recommend a setting.
- Shipped with **zero runtime dependencies and no build step** — five charts hand-written in canvas — plus English / Traditional Chinese across 168 strings, including canvas labels.
- Deployed via GitHub Actions staging an explicit file allowlist, and covered it with **123 assertions across five suites plus 33 browser and 23 end-to-end checks** after an audit found four silently wrong metrics.

### Handwritten-digit recognition on an Arduino (int8 on-device inference)

Python · TensorFlow · AVR C++

Course project · github.com/lyimou/arduino-mnist-inference

- Trained a Keras classifier on binarised 14×14 MNIST, then quantised the weights to **int8 with per-matrix scales** and emitted ~1.7 KB of PROGMEM arrays so the network fits an 8-bit AVR's flash.
- Ran all inference on the microcontroller over a 115200-baud serial protocol (READY handshake, 196-character payload) with buffer resets and error recovery, measuring **75.9% accuracy over 1,500 held-out samples**.

### Spire Climb — browser deck-building roguelike

Vanilla JavaScript · CSS3

github.com/lyimou/spire-climb-web

- Built a turn-based card battler with no framework and no build step: procedural map, three heroes, energy/block/damage resolution and draw—hand—discard piles, with cards defined in JSON apart from the game logic.
- Audited it with a headless Playwright harness driving the real page, finding and fixing 13 state bugs — including a shuffle that silently reset the draw pile (card totals drifted 9 → 13 → 11).

### C++ data-structures implementations

C++11 · templates · RAII

Course project — implementation excerpts; repository private

- Hand-wrote shared\_ptr, weak\_ptr and their control block, plus a 435-line **trie** for prefix search and key-value **BST** templates under an **LRU cache**.

## SKILLS

- Languages:** C++ (C++11, templates, RAII, smart pointers), Python, JavaScript (ES modules), HTML/CSS
- Data structures & algorithms:** hash tables, tries, binary search trees, LRU caches, graphs; complexity analysis
- ML & hardware:** TensorFlow/Keras, NumPy, int8 quantisation, on-device inference; Arduino / AVR, UART serial
- Web & tooling:** Canvas 2D, Web APIs (Pointer Lock, localStorage), i18n, Git, GitHub Actions, Node.js, Linux